

Experiment - 03 & 04

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(A):

You are given with the Employee schema; the task is to find the highest score for each respective department.

For eg: department D1 have the highest score as 5.28. Hence in the output, final updated score for D1 should be 5.28. Same with the other departments i.e., D2, D3 and D4.

EID	DEPT	SCORES
1	D1	1
2	D1	5.28
3	D1	4
4	D2	8
5	D1	2.5
6	D2	7
7	D3	9
8	D4	10.2

```
UPDATE Employee e
SET Scores = (
    SELECT MAX(Scores)
    FROM Employee
    WHERE Dept = e.Dept
);
```

(B):

A company maintains employee salary records for different departments. The management wants to prepare a report containing employees whose salaries fall among the **top three highest distinct salary values** within their respective departments. If multiple employees receive the same qualifying salary, all such employees must be included in the final report. Display the department name, employee name, and salary for every qualifying employee.

```
SELECT
D.NAME AS DEPARTMENT,
E.NAME AS EMPLOYEE,
E.SALARY AS SALARY
FROM
EMPLOYEES AS E
JOIN
DEPARTMENT AS D
ON
E.departmentId = D.id
WHERE
(
    SELECT COUNT(DISTINCT E2.SALARY)
    FROM EMPLOYEES AS E2
    WHERE
    E2.departmentId = E.departmentId
    AND
    E2.SALARY > E.SALARY
) < 3
ORDER BY
D.NAME;
```

(C): SALES ANALYSIS PART 1

A company maintains records of products sold by different salespersons. The management wants to prepare a sales performance report by identifying the salesperson(s) who generated the highest overall sales revenue. If multiple salespersons have the same highest total sales amount, all of them must be included in the final report. Return the seller ID(s) satisfying this condition.

```
SELECT seller_id
FROM orders
GROUP BY seller_id
HAVING SUM(amount) = (
    SELECT MAX(total_sales)
    FROM (
        SELECT seller_id, SUM(amount) AS total_sales
        FROM orders
        GROUP BY seller_id
    ) AS t
);
```

(A)

(i) You are given a table `tbl_happiness` that contains the ranking of happiness for different countries. Your task is to retrieve all the records from the table **while ensuring that "India" and "Sri Lanka" appear at the top, followed by the rest of the countries in their original ranking order.**

(B)

(ii) Make all the rankings in descending order except **India** and **Sri Lanka**.

S.No	Rankings	Country
1	1	Finland
2	2	Denmark
3	3	Iceland
4	4	Israel
5	5	Netherlands
6	6	Sweden
7	7	Norway
8	126	India
9	128	Sri Lanka

```
SELECT
    T.rankings,
    T.country
FROM tbl_happiness AS T
ORDER BY
    CASE
        WHEN T.country = 'India' THEN 0
        WHEN T.country = 'Sri Lanka' THEN 0
        ELSE T.rankings
    END;
```

```
SELECT
    T.rankings,
    T.country
FROM tbl_happiness AS T
ORDER BY
    CASE
        WHEN T.country = 'India' THEN 0
        WHEN T.country = 'Sri Lanka' THEN 0
        ELSE T.rankings DESC
    END;
```

(C)

A company maintains employee information in a single table where each employee may report to another employee acting as their manager. Prepare a report that displays the details of every employee who supervises at least one other employee. The report should include the manager's employee ID, name, total number of direct reports, and the average age of those reporting employees. The average age should be rounded to the nearest whole number and the final result should be sorted by employee ID.

employee_id	name	reports_to	age
9	Hercy	NULL	43
6	Alice	9	41
4	Bob	9	36
2	Winston	NULL	37

```
SELECT
    m.employee_id,
    m.name,
    COUNT(e.employee_id) AS reports_count,
    ROUND(AVG(e.age)) AS average_age
FROM Employees AS m
JOIN Employees AS e
    ON m.employee_id = e.reports_to
GROUP BY
    m.employee_id,
    m.name
ORDER BY
    m.employee_id;
```